

# Genetic Pathway Reference: Strength, Hypertrophy and Power

Educational, shareable document · GRCh38 coordinates

03-10-2026

Generic reference: contains no personal genetic data, medication or supplementation of any individual.

Educational material. Not medical advice; clinical decisions should be discussed with a treating physician.

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## 1. How to use this reference

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This reference summarises the biological pathways and best-studied genes linked to muscle strength, hypertrophy and power, with the SNPs (single-nucleotide variants) that can be looked up in a genetic file. It is meant for physicians, coaches, researchers or anyone who wants to interpret a genotype in context.

- **Alleles:** always on the + strand of the GRCh38 reference genome, unless marked "gene strand" (genes read on the – strand, where the literature uses the complementary letters).
- **Effect/risk allele:** the allele whose effect is described in the evidence column. In several genes there is no "bad allele": there are trade-offs (for example, power versus endurance).
- **Evidence level:** Strong = replicated in GWAS or meta-analyses with a clear effect; Moderate = consistent association of modest effect; Weak = small candidate-gene studies or mechanistic only; Mixed = contradictory results.
- **ClinVar and PharmGKB:** ClinVar pathogenicity status and the highest PharmGKB clinical annotation level (1A = clinical guideline; 4 = weak evidence).
- **Effect size:** almost all effects are small. Single genes explain a few per cent of the variation in strength or mass; training, nutrition, sleep and age weigh much more.
- **Frequency/effect data:** many candidate-gene studies used small samples from a single population; for other reference populations, frequency/effect data are weak or absent.

## 2. Pathway biology

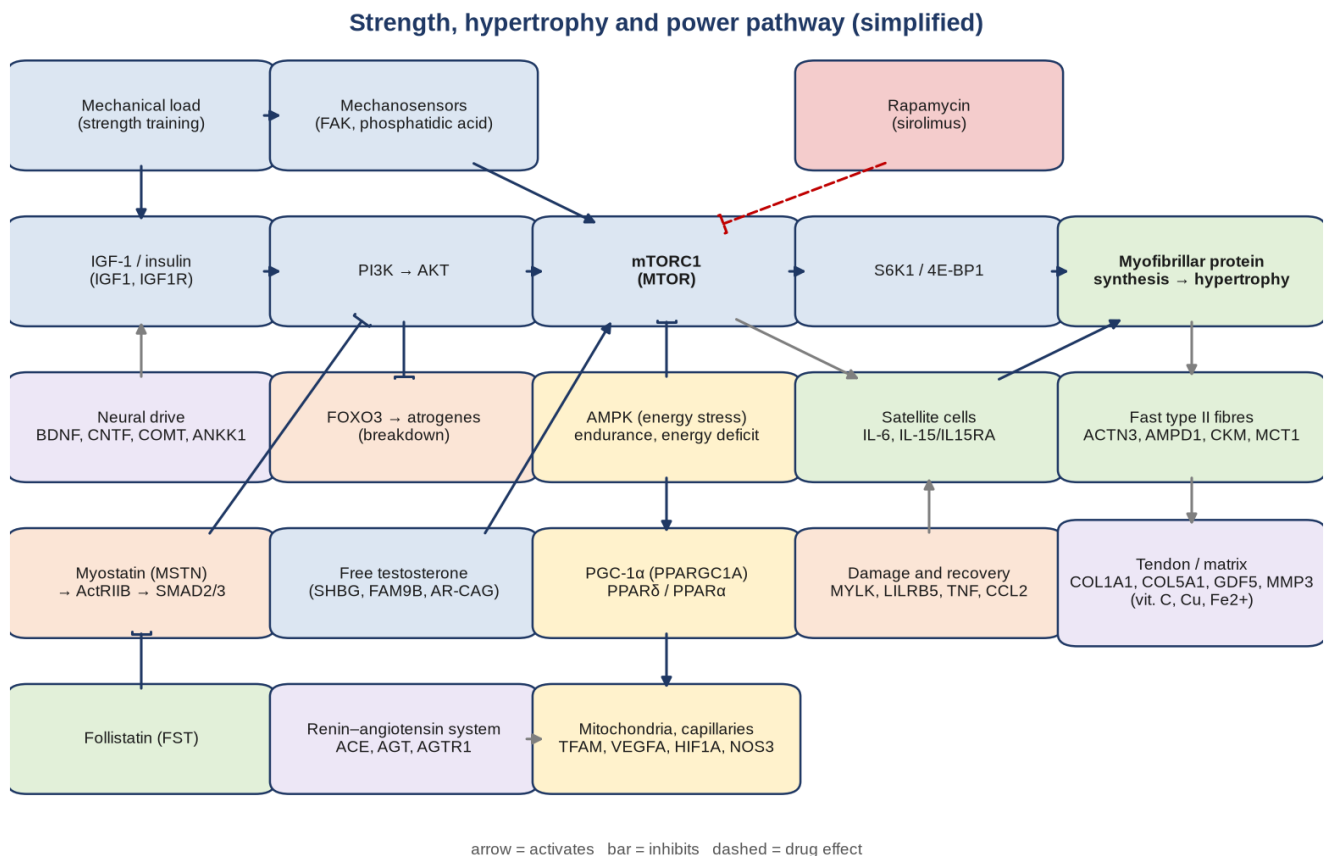


Figure: simplified scheme. Mechanical load, IGF-1 and amino acids activate mTORC1, which switches on myofibrillar protein synthesis; myostatin, AMPK and rapamycin brake this pathway.

### From stimulus to bigger muscle

Mechanical tension is the main stimulus for hypertrophy. It is sensed by membrane and cytoskeletal mechanosensors (focal adhesions, phosphatidic acid production) that activate **mTORC1** partly independently of IGF-1. mTORC1 phosphorylates **S6K1** and **4E-BP1** and raises protein translation for 24–48 h after each session. **IGF-1** and insulin reinforce the signal through the PI3K–AKT axis, which also switches off **FOXO** factors (and with them the breakdown genes, the "atrogenes"). Amino acids, especially leucine, activate mTORC1 at the lysosome. The net balance between synthesis and breakdown over weeks decides whether muscle grows (Roberts 2023).

### Brakes and interference

**Myostatin** (MSTN), via the ActRIIB receptor and SMAD2/3, brakes AKT and mTORC1; its total loss gives extreme hypertrophy (Schuelke 2004). **Follistatin** neutralises myostatin. **AMPK**, activated by energy shortage (prolonged endurance exercise, energy deficit), inhibits mTORC1 and activates **PGC-1α**, which pushes towards mitochondria and capillaries: this is the biological basis of "interference" between endurance and strength training. **Rapamycin** directly inhibits mTORC1 and, in humans, blocks the contraction-induced rise in protein synthesis (Drummond 2009).

### Myofibrillar and sarcoplasmic hypertrophy

In **myofibrillar hypertrophy**, contractile proteins and force per cross-sectional area increase. In **sarcoplasmic hypertrophy**, cytoplasmic volume (glycogen, enzymes, water) rises, giving size without a proportional gain in strength; it appears more common with high volumes and moderate loads, but it is hard to measure and the evidence is limited (Roberts 2020). No validated SNPs separate the two; lean-mass genetics (GWAS of tens to hundreds of thousands of people: Zillikens 2017; Pei 2020) measures the combined result.

### Strength, power and the nervous system

Maximal **strength** depends on muscle mass, architecture and neural drive (motor-unit recruitment, firing rate, coordination). In the first weeks of training, most of the gain is neural. **Power** (force × velocity) depends mainly on fast type II fibres, rapid ATP

regeneration (phosphocreatine and AMP deaminase) and tendon stiffness. Grip-strength GWAS (Willems 2017; Tikkanen 2018) show that strength is polygenic, with loci linked to nervous-system development, muscle structure and metabolism.

## Trainability

With the same programme, size and strength gains vary enormously (from almost nothing to over 50% in 12 weeks; Hubal 2005), and the VO2max training response clusters in families (Bouchard HERITAGE; Bouchard 2011). Single genes explain little of this variation; volume, frequency, load progression, protein, sleep and available energy are the modulators with the largest effect (Schoenfeld 2017; Morton 2018; Murphy 2022).

## Tendons, heart, metabolism and hormones

**Tendon** transmits force and adapts more slowly than muscle; the mismatch between a fast-growing muscle and a tendon that has not yet adapted is a classic cause of tendinopathy. The **cardiovascular system** supplies oxygen and removes metabolites; endothelial nitric oxide controls flow to active muscle. Glucose and fat **metabolism** set the fuel and body composition. **Hormones** (free testosterone, IGF-1, vitamin D) modulate protein synthesis and satellite cells.

## Relevant pharmacology (informational)

mTOR inhibitors (sirolimus, everolimus), statins, fluoroquinolones, oral retinoids and incretin agonists have known effects on muscle or tendon (Drummond 2009; Cooper-DeHoff 2022; Alves 2019; Marson 2022; Look 2025). Sirolimus and tacrolimus are **narrow-therapeutic-index immunosuppressants**: any pharmacogenomic finding (CYP3A5, CYP3A4, ABCB1) is informational only and requires review by the prescribing physician and, when indicated, blood drug-level monitoring.

### 3. Functional category map

Ten functional categories, each split into subcategories. The same gene may act in more than one; it appears under its main effect.

| Category                                      | Subcategories                                                                                                             | Genes                                                                        |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1. Fibre type, power and speed                | Fast-fibre structure; Rapid anaerobic energy; Muscle renin–angiotensin system; Hypoxia signal, PPAR $\alpha$ and myokines | ACTN3, AMPD1, GALNTL6, ACE, AGT, HIF1A, PPARA, IL6                           |
| 2. Myofibrillar hypertrophy and lean mass     | IGF-1 signal and thyroid axis; Growth brakes (myostatin and follistatin)                                                  | IGF1, IGF1R, TRHR, MSTN, FST                                                 |
| 3. Sarcoplasmic hypertrophy and muscle energy | Mitochondrial biogenesis; Lactate and phosphocreatine                                                                     | PPARGC1A, TFAM, SLC16A1, CKM                                                 |
| 4. Strength and trainability                  | Neurotrophic factors and myokines; Metabolic training response; Grip-strength GWAS loci                                   | CNTF, IL15RA, PPARG, ACSL1, TGFA, POLD3, HOXB3, GLIS1, PEX14, KANSL1, LRPPRC |
| 5. Muscle damage, recovery and inflammation   | Contraction and damage; Creatine kinase (CK); Inflammation and redox; Statin myopathy                                     | MYLK, LILRB5, TNF, CCL2, SOD2, SLC01B1                                       |
| 6. Tendons and connective tissue              | Collagens; Matrix growth and remodelling; Injury GWAS locus                                                               | COL5A1, COL1A1, COL3A1, GDF5, MMP3, MMP7                                     |
| 7. Cardiovascular adaptation and oxygen       | Nitric oxide and endothelium; Capillaries; Heart and renin–angiotensin system; Iron and oxygen transport                  | NOS3, EDN1, VEGFA, ADRB1, AGTR1, HFE                                         |
| 8. Glucose, fat and fuel                      | Insulin and glucose; Adiposity and appetite; Lipolysis and thermogenesis                                                  | TCF7L2, MTNR1B, PPARG, FTO, MC4R, ADRB3, UCP2                                |
| 9. Brain, motivation and motor control        | Motor plasticity; Dopamine and reward; Exercise behaviour                                                                 | BDNF, COMT, ANKK1, CADM2                                                     |
| 10. Hormones, mTOR and rapamycin              | Androgens and SHBG; Vitamin D; mTOR and longevity; Rapamycin pharmacogenomics                                             | SHBG, FAM9B, SRD5A2, VDR, GC, MTOR, FOXO3, CYP3A5, CYP3A4, ABCB1             |

## 4. Fibre type, power and speed

Power depends on fast (type II) fibres, which contract quickly and generate high force in little time. These genes affect the structure of those fibres, rapid ATP regeneration and the balance between fast and slow fibres.

### Fast-fibre structure

| Gene         | SNP · variant                 | Effect/risk allele | Functional consequence                                                                                                                                                                                  | Cofactors                              | Evidence (GWAS p, effect)                                                                                                                                                                                                                                                | ClinVar · PharmGKB  |
|--------------|-------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>ACTN3</b> | rs1815739 · R577X (c.1729C>T) | T (X, stop codon)  | R577X stop codon: the X allele removes $\alpha$ -actinin-3, a Z-line protein of fast (type II) fibres. XX $\approx$ 18% of the reference population; XX muscle shifts towards a more oxidative profile. | None (structural protein; binds actin) | RR/RX enriched in sprint and power athletes; XX linked to more damage after eccentric exercise. Explains only ~2–3% of strength variance; in non-athletes the evidence is MIXED. <b>Level: Mixed.</b> (Yang 2003; MacArthur 2007; Clarkson 2005; Del Coso 2017; Ma 2013) | Benign · PharmGKB — |

### Rapid anaerobic energy

| Gene           | SNP · variant            | Effect/risk allele  | Functional consequence                                                                                    | Cofactors                                       | Evidence (GWAS p, effect)                                                                                                                                         | ClinVar · PharmGKB                                            |
|----------------|--------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>AMPD1</b>   | rs17602729 · C34T (Q12X) | A (34T, stop codon) | Q12X: early stop codon; homozygotes (~2%) lack muscle AMP deaminase (ATP regeneration in intense effort). | Zn <sup>2+</sup> (metalloenzyme)                | Carriers show a poorer cardiorespiratory training response (HERITAGE family study); homozygotes have fatigue and cramps. <b>Level: Moderate.</b> (Rico-Sanz 2003) | Conflicting classifications; Other · PharmGKB 3 (regadenoson) |
| <b>GALNTL6</b> | rs558129 · intronic      | T (gene strand)     | Intronic; associated with anaerobic (power) performance.                                                  | Mn <sup>2+</sup> (GalNAc-T glycosyltransferase) | Candidate-gene studies; no GWAS. <b>Level: Weak.</b> (Díaz Ramírez 2020)                                                                                          | No record · PharmGKB —                                        |

### Muscle renin–angiotensin system

| Gene       | SNP · variant                         | Effect/risk allele    | Functional consequence                                                                                                                                        | Cofactors                                           | Evidence (GWAS p, effect)                                                                                                                                                                                                                                                       | ClinVar · PharmGKB               |
|------------|---------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>ACE</b> | rs4343 · A2350G (I/D proxy; G tags D) | G (tags the D allele) | Synonymous variant in strong linkage with the intron-16 Alu insertion/deletion (I/D). The D allele gives higher angiotensin-converting enzyme (ACE) activity. | Zn <sup>2+</sup> (metalloprotease), Cl <sup>-</sup> | ACE-activity GWAS: p $3 \times 10^{-25}$ , 16% of variance. D linked to strength and power gains and I to endurance; replication is inconsistent: MIXED evidence. <b>Level: Mixed.</b> (Rigat 1990; Montgomery 1998; Folland 2000; Williams 2005; Chung 2010; Puthucherry 2011) | Benign · PharmGKB 3 (sildenafil) |
| <b>AGT</b> | rs699 · M235T (c.704T>C)              | G (235T)              | M235T: the 235T allele is associated with higher circulating angiotensinogen and renin–angiotensin activity.                                                  | None (renin substrate)                              | More frequent in power than endurance athletes; small studies. <b>Level: Weak.</b> (Zarebska 2013)                                                                                                                                                                              | Benign · PharmGKB 3 (irbesartan) |

ACE note: the 287-bp Alu insertion/deletion is not a SNP and is not on chips; rs4343 is used as a strongly linked proxy marker. Direct confirmation requires PCR or sequencing.

### Hypoxia signal, PPAR $\alpha$ and myokines

| Gene         | SNP · variant               | Effect/risk allele    | Functional consequence                                                                             | Cofactors                                                                                                                        | Evidence (GWAS p, effect)                                                                    | ClinVar · PharmGKB                                            |
|--------------|-----------------------------|-----------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>HIF1A</b> | rs11549465 · P582S (C1772T) | T (Ser582, advantage) | P582S: Ser582 makes HIF-1 $\alpha$ more stable in normoxia, with more glycolysis and angiogenesis. | Fe <sup>2+</sup> , 2-oxoglutarate, O <sub>2</sub> and vitamin C (cofactors of the PHD hydroxylases that degrade HIF-1 $\alpha$ ) | More frequent in strength athletes; limited replication. <b>Level: Weak.</b> (Gabbasov 2013) | Conflicting classifications · PharmGKB 3 (axitinib/sorafenib) |

|              |                             |           |                                                                                                                           |                                      |                                                                                                                                    |                                                  |
|--------------|-----------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>PPARA</b> | rs4253778 ·<br>intron 7 G/C | C (power) | Intron 7 G/C: the C allele lowers PPAR $\alpha$ expression, with less fat oxidation and more fast fibres and hypertrophy. | Ligands: fatty acids and eicosanoids | C linked to power and G to endurance; MIXED replication. <b>Level: Mixed.</b> (Ahmetov 2006; Ahmetov 2013)                         | No record ·<br>PharmGKB 3 (Beta Blocking Agents) |
| <b>IL6</b>   | rs1800795 ·<br>-174 G>C     | G (-174G) | Promoter -174 G>C: the G allele is associated with higher exercise-induced IL-6 (an anabolic and inflammatory myokine).   | —                                    | G enriched in power athletes in one study, not replicated in another: MIXED evidence. <b>Level: Mixed.</b> (Ruiz 2010; Eynon 2011) | Other; Risk factor ·<br>PharmGKB 3 (fenofibrate) |



## 5. Myofibrillar hypertrophy and lean mass

Myofibrillar hypertrophy is the growth of contractile proteins (actin, myosin). The engine is mTORC1, activated by mechanical load, IGF-1/insulin and amino acids; myostatin is the brake. Lean mass is highly polygenic: polygenic scores say more than any single SNP.

### IGF-1 signal and thyroid axis

| Gene         | SNP · variant               | Effect/risk allele            | Functional consequence                                                                                                                               | Cofactors                                       | Evidence (GWAS p, effect)                                                                                                                      | ClinVar · PharmGKB     |
|--------------|-----------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>IGF1</b>  | rs35767 · promoter -383 C>T | T in the gene (A on + strand) | IGF1 promoter -383 C>T: the T allele is associated with higher circulating IGF-1 in some studies; the + strand G allele with higher fasting insulin. | Growth hormone, protein and energy intake, zinc | Fasting-insulin GWAS (G allele): p $2 \times 10^{-9}$ . Effect on IGF-1 inconsistent. <b>Level: Moderate.</b> (GWAS Catalog)                   | No record · PharmGKB — |
| <b>IGF1R</b> | rs2229765 · E1043E (G3174A) | A                             | Synonymous IGF-1 receptor variant; the A allele is associated with lower free IGF-1 and, in some studies, with longevity.                            | ATP, Mg <sup>2+</sup> (tyrosine kinase)         | Candidate-gene studies. <b>Level: Weak.</b>                                                                                                    | Benign · PharmGKB —    |
| <b>TRHR</b>  | rs7832552 · intronic        | T (more lean mass)            | Intronic in the TRH receptor; the T allele is associated with more lean mass.                                                                        | TRH signal (thyroid axis)                       | Lean-mass GWAS: p $4 \times 10^{-10}$ , $\beta$ +0.10 units per allele; limited replication in larger GWAS. <b>Level: Moderate.</b> (Liu 2009) | No record · PharmGKB — |

### Growth brakes (myostatin and follistatin)

| Gene        | SNP · variant                | Effect/risk allele | Functional consequence                                                                                                                          | Cofactors                                                                          | Evidence (GWAS p, effect)                                                                                         | ClinVar · PharmGKB     |
|-------------|------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>MSTN</b> | rs1805086 · K153R (c.458A>G) | C (153R)           | K153R in myostatin (the brake on muscle growth); total loss of myostatin gives extreme hypertrophy, but this common variant has a small effect. | Activated by furin and BMP-1/tolloid (Zn <sup>2+</sup> -dependent metalloprotease) | Inconsistent association with power; the R allele is uncommon. <b>Level: Weak.</b> (Schuelke 2004; Santiago 2011) | Benign · PharmGKB —    |
| <b>FST</b>  | rs3797297 · intronic         | T                  | Intronic in follistatin (inhibits myostatin and activin).                                                                                       | —                                                                                  | Association with strength in small studies; no GWAS. <b>Level: Weak.</b>                                          | No record · PharmGKB — |

## 6. Sarcoplasmic hypertrophy and muscle energy

Sarcoplasmic hypertrophy is expansion of the cytoplasm (glycogen, enzymes, mitochondria, water) without a proportional rise in myofibrils. It is real but hard to measure, and **no validated SNP specific to it exists**; genes of muscle energy metabolism are used as a proxy.

### Mitochondrial biogenesis

| Gene            | SNP · variant                 | Effect/risk allele | Functional consequence                                                                                                        | Cofactors                                                             | Evidence (GWAS p, effect)                                                                       | ClinVar · PharmGKB  |
|-----------------|-------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------|
| <b>PPARGC1A</b> | rs8192678 · G482S (c.1444G>A) | T (Ser482)         | G482S: Ser482 lowers PGC-1 $\alpha$ co-activator activity, with less mitochondrial biogenesis, capillaries and type I fibres. | Activated by phosphorylation (AMPK, p38) — calcium and energy signals | Gly/Gly enriched in elite endurance athletes. <b>Level: Moderate.</b> (Lucia 2005; Stefan 2007) | Benign · PharmGKB — |
| <b>TFAM</b>     | rs1937 · S12T                 | C (Thr12)          | S12T in mitochondrial transcription factor A (copying and maintenance of mitochondrial DNA).                                  | — (binds mitochondrial DNA)                                           | Associations with VO2max in small studies. <b>Level: Weak.</b>                                  | Benign · PharmGKB — |

### Lactate and phosphocreatine

| Gene           | SNP · variant               | Effect/risk allele                   | Functional consequence                                                                            | Cofactors                                           | Evidence (GWAS p, effect)                                                                                             | ClinVar · PharmGKB     |
|----------------|-----------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>SLC16A1</b> | rs1049434 · D490E (T1470A)  | T (gene strand, 1470T)               | D490E in the MCT1 lactate transporter: the T allele reduces lactate transport out of fast fibres. | H <sup>+</sup> gradient (lactate/proton symport)    | More lactate accumulation during resistance circuits in T carriers; small studies. <b>Level: Weak.</b> (Cupeiro 2012) | Benign · PharmGKB —    |
| <b>CKM</b>     | rs8111989 · 3' UTR NcoI A/G | A/G (NcoI; no consensus risk allele) | 3' UTR of muscle creatine kinase (CK-MM), which regenerates ATP from phosphocreatine.             | Mg <sup>2+</sup> -ATP; creatine and phosphocreatine | Linked to the CK response to exercise and to VO2max; MIXED results. <b>Level: Mixed.</b> (Heled 2007)                 | No record · PharmGKB — |

## 7. Strength and trainability

Trainability (how much is gained from the same training) varies widely: with strength training, size gains range from almost zero to over 50%. Part of this variation runs in families, but almost no single gene predicts it well; strength GWAS loci have small effects that add up.

### Neurotrophic factors and myokines

| Gene          | SNP · variant                           | Effect/risk allele   | Functional consequence                                                                 | Cofactors | Evidence (GWAS p, effect)                                                                                                    | ClinVar · PharmGKB     |
|---------------|-----------------------------------------|----------------------|----------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>CNTF</b>   | rs1800169 · G-A (null allele, splicing) | A (null allele)      | Splicing error that abolishes CNTF (motor-nerve neurotrophic factor); homozygotes ~2%. | —         | Associated with strength and muscle quality in one candidate-gene study. <b>Level: Weak.</b> (Roth 2001)                     | Benign · PharmGKB —    |
| <b>IL15RA</b> | rs2296135 · 3' UTR A/C                  | A (more hypertrophy) | 3' UTR of the IL-15 receptor $\alpha$ (anabolic myokine).                              | —         | The A allele explained part of the hypertrophic response to training in one small study. <b>Level: Weak.</b> (Riechman 2004) | No record · PharmGKB — |

### Metabolic training response

| Gene         | SNP · variant            | Effect/risk allele | Functional consequence                                                                                                                                                   | Cofactors                         | Evidence (GWAS p, effect)                                                                                   | ClinVar · PharmGKB     |
|--------------|--------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------|
| <b>PPARD</b> | rs2267668 · intronic A>G | G (lower response) | Intronic in PPAR $\delta$ (fat oxidation and oxidative fibres): the G allele is associated with smaller gains in aerobic capacity and insulin sensitivity with training. | Ligands: fatty acids              | Lifestyle intervention; limited replication. <b>Level: Weak.</b> (Stefan 2007)                              | No record · PharmGKB — |
| <b>ACSL1</b> | rs6552828 · intronic     | A (+ strand)       | Intronic in ACSL1 (activation of fatty acids for oxidation).                                                                                                             | ATP, coenzyme A, Mg <sup>2+</sup> | Predictor of the VO <sub>2</sub> max response to standardised training. <b>Level: Weak.</b> (Bouchard 2011) | No record · PharmGKB — |

### Grip-strength GWAS loci

| Gene          | SNP · variant                | Effect/risk allele | Functional consequence                    | Cofactors                       | Evidence (GWAS p, effect)                                                                                                                                               | ClinVar · PharmGKB     |
|---------------|------------------------------|--------------------|-------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>TGFA</b>   | rs958685 · grip GWAS locus   | A (more strength)  | TGFA locus (growth factor).               | —                               | Grip GWAS: p $5 \times 10^{-13}$ , $\beta$ +0.157 per allele; also muscle weakness after 60 (p $7 \times 10^{-9}$ ). <b>Level: Moderate.</b> (Willems 2017; Jones 2021) | No record · PharmGKB — |
| <b>POLD3</b>  | rs72979233 · grip GWAS locus | A (more strength)  | POLD3 locus (DNA replication).            | —                               | Grip GWAS: p $5 \times 10^{-13}$ , $\beta$ +0.175 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                    | No record · PharmGKB — |
| <b>HOXB3</b>  | rs2288278 · grip GWAS locus  | A (more strength)  | HOXB3 locus (development).                | —                               | Grip GWAS: p $4 \times 10^{-12}$ , $\beta$ +0.157 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                    | No record · PharmGKB — |
| <b>GLIS1</b>  | rs4926611 · grip GWAS locus  | C (more strength)  | GLIS1 locus (transcription factor).       | Zn <sup>2+</sup> (zinc fingers) | Grip GWAS: p $5 \times 10^{-12}$ , $\beta$ +0.156 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                    | No record · PharmGKB — |
| <b>PEX14</b>  | rs6687430 · grip GWAS locus  | G (more strength)  | PEX14 locus (peroxisomes).                | —                               | Grip GWAS: p $6 \times 10^{-11}$ , $\beta$ +0.142 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                    | No record · PharmGKB — |
| <b>KANSL1</b> | rs80103986 · grip GWAS locus | A (more strength)  | KANSL1 locus (17q21 inversion region).    | —                               | Grip GWAS: p $1 \times 10^{-9}$ , $\beta$ +0.171 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                     | No record · PharmGKB — |
| <b>LRPPRC</b> | rs10186876 · grip GWAS locus | A (more strength)  | LRPPRC locus (mitochondrial translation). | —                               | Grip GWAS: p $1 \times 10^{-10}$ , $\beta$ +0.147 per allele. <b>Level: Moderate.</b> (Willems 2017)                                                                    | No record · PharmGKB — |

Grip loci have very small per-allele effects ( $\approx 0.14$ – $0.18$  grip-strength units); they are only useful summed into a polygenic score.

## 8. Muscle damage, recovery and inflammation

Eccentric exercise causes micro-injury; recovery involves controlled inflammation (macrophages, cytokines) and satellite cells. These genes affect damage, markers (CK) and the inflammatory response, and include the pharmacogenomics of statin myopathy.

### Contraction and damage

| Gene        | SNP · variant               | Effect/risk allele     | Functional consequence                                                                           | Cofactors                                           | Evidence (GWAS p, effect)                                                                                                 | ClinVar · PharmGKB                |
|-------------|-----------------------------|------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>MYLK</b> | rs28497577 · C37885A (P21H) | A in the gene (37885A) | C37885A (P21H) in myosin light-chain kinase (phosphorylates myosin and tunes contractile force). | Ca <sup>2+</sup> /calmodulin, ATP, Mg <sup>2+</sup> | Carriers had higher CK and more strength loss after eccentric exercise; small study. <b>Level: Weak.</b> (Clarkson 2005b) | Benign/likely benign · PharmGKB — |
| <b>MYLK</b> | rs2700352 · C49T (P147S)    | T in the gene (49T)    | C49T (P147S) in the same kinase (MYLK).                                                          | Ca <sup>2+</sup> /calmodulin, ATP, Mg <sup>2+</sup> | Associated with exercise-induced muscle damage; small study. <b>Level: Weak.</b> (Clarkson 2005b)                         | Benign · PharmGKB —               |

### Creatine kinase (CK)

| Gene          | SNP · variant                   | Effect/risk allele | Functional consequence                                                          | Cofactors | Evidence (GWAS p, effect)                                                                                                               | ClinVar · PharmGKB  |
|---------------|---------------------------------|--------------------|---------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>LILRB5</b> | rs12975366 · D247G (T = Asp247) | T (Asp247)         | D247G in the immune receptor LILRB5: the Asp247 allele gives higher resting CK. | —         | CK GWAS: p 2×10 <sup>-144</sup> ; Asp247 linked to myalgia and statin intolerance. <b>Level: Moderate.</b> (Siddiqui 2017; Tornio 2023) | Benign · PharmGKB — |

### Inflammation and redox

| Gene        | SNP · variant         | Effect/risk allele | Functional consequence                                                                                                                                           | Cofactors        | Evidence (GWAS p, effect)                                                                                       | ClinVar · PharmGKB                                                       |
|-------------|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>TNF</b>  | rs1800629 · -308 G>A  | A (-308A)          | Promoter -308 G>A: the A allele is associated with higher TNF-α.                                                                                                 | —                | Inflammation and recovery: MIXED studies; PharmGKB 2B (etanercept). <b>Level: Mixed.</b>                        | Drug response · PharmGKB 2B (etanercept)                                 |
| <b>CCL2</b> | rs1024611 · -2518 A>G | G (-2518G)         | Promoter -2518 A>G: the G allele is associated with higher MCP-1 (recruits macrophages to damaged muscle).                                                       | —                | Linked to more damage and strength loss after eccentric exercise in candidate-gene studies. <b>Level: Weak.</b> | Uncertain significance · PharmGKB —                                      |
| <b>SOD2</b> | rs4880 · V16A         | G (Ala16)          | V16A: Ala16 enters mitochondria more efficiently, converting more superoxide into H <sub>2</sub> O <sub>2</sub> (an adaptive signal and, in excess, a stressor). | Mn <sup>2+</sup> | MIXED results for performance and disease. <b>Level: Mixed.</b>                                                 | Conflicting classifications; Risk factor · PharmGKB 3 (cyclophosphamide) |

### Statin myopathy

| Gene           | SNP · variant                | Effect/risk allele | Functional consequence                                                                                             | Cofactors | Evidence (GWAS p, effect)                                                                                                                                    | ClinVar · PharmGKB                                              |
|----------------|------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>SLCO1B1</b> | rs4149056 · V174A (c.521T>C) | C (Ala174)         | V174A in the hepatic transporter OATP1B1: the C allele reduces liver uptake of statins and raises muscle exposure. | —         | Simvastatin myopathy: OR ~4.5 per allele (GWAS, p 4×10 <sup>-9</sup> ); CPIC guideline; PharmGKB 1A. <b>Level: Strong.</b> (SEARCH 2008; Cooper-DeHoff 2022) | Drug response · PharmGKB 1A (statins, incl. pitavastatin; CPIC) |

## 9. Tendons and connective tissue

Tendon adapts to load more slowly than muscle (months rather than weeks). Collagen synthesis needs vitamin C, iron and 2-oxoglutarate (hydroxylases) and copper (lysyl oxidase), and responds to mTORC1.

### Collagens

| Gene          | SNP · variant                        | Effect/risk allele                        | Functional consequence                                                                                                             | Cofactors                                                                             | Evidence (GWAS p, effect)                                                                                                                                                            | ClinVar · PharmGKB                                             |
|---------------|--------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>COL5A1</b> | rs12722 · BstUI 3' UTR C>T           | T                                         | BstUI 3' UTR of collagen V (which sets collagen I fibril diameter): the T allele changes mRNA stability and gives stiffer tendons. | Vitamin C, Fe <sup>2+</sup> and 2-oxoglutarate (hydroxylases); copper (lysyl oxidase) | TT associated with Achilles tendinopathy and reduced range of motion; MIXED replication across populations. <b>Level: Moderate.</b> (Mokone 2006; Collins 2009; Brown 2017)          | Benign · PharmGKB —                                            |
| <b>COL1A1</b> | rs1800012 · Sp1 G>T (T allele = "s") | T in the gene ("s" allele, A on + strand) | Intron-1 Sp1 site of collagen I: the s allele raises α1 chains (altered α1:α2 ratio).                                              | Vitamin C, Fe <sup>2+</sup> , 2-oxoglutarate; copper                                  | ss protective against ligament and tendon ruptures, but the s allele is linked to lower bone density; MIXED effect. <b>Level: Mixed.</b> (Posthumus 2009; Collins 2010; Gibbon 2020) | Association; Benign/likely benign · PharmGKB 3 (valproic acid) |
| <b>COL3A1</b> | rs1800255 · A698T                    | A (698T)                                  | A698T in collagen III (healing tendon, vessels).                                                                                   | Vitamin C, Fe <sup>2+</sup> , 2-oxoglutarate; copper                                  | Ligament injuries in small studies. <b>Level: Weak.</b>                                                                                                                              | Benign · PharmGKB —                                            |

### Matrix growth and remodelling

| Gene        | SNP · variant             | Effect/risk allele            | Functional consequence                                                                         | Cofactors                           | Evidence (GWAS p, effect)                                                                  | ClinVar · PharmGKB     |
|-------------|---------------------------|-------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|------------------------|
| <b>GDF5</b> | rs143383 · +104T/C 5' UTR | T in the gene (A on + strand) | +104T/C in the GDF5 5' UTR: the T allele lowers GDF5 expression (cartilage and tendon growth). | —                                   | Osteoarthritis (meta-analyses) and tendinopathy (smaller studies). <b>Level: Moderate.</b> | No record · PharmGKB — |
| <b>MMP3</b> | rs679620 · E45K           | E45K (no consensus)           | E45K in metalloprotease MMP3 (tendon matrix remodelling).                                      | Zn <sup>2+</sup> , Ca <sup>2+</sup> | Achilles tendinopathy in small studies, interacting with COL5A1. <b>Level: Weak.</b>       | Benign · PharmGKB —    |

### Injury GWAS locus

| Gene        | SNP · variant                          | Effect/risk allele | Functional consequence       | Cofactors | Evidence (GWAS p, effect)                                                                                   | ClinVar · PharmGKB     |
|-------------|----------------------------------------|--------------------|------------------------------|-----------|-------------------------------------------------------------------------------------------------------------|------------------------|
| <b>MPP7</b> | rs1249269 · Achilles injury GWAS locus | T                  | MPP7 locus (cell junctions). | —         | Achilles tendon injury GWAS: OR 1.09 per allele, p 1×10 <sup>-10</sup> . <b>Level: Moderate.</b> (Kim 2021) | No record · PharmGKB — |

## 10. Cardiovascular adaptation and oxygen

Strength and power training also depend on blood flow (exercise hyperaemia), capillarisation and oxygen transport. Endothelial nitric oxide (eNOS) is the central vasodilator.

### Nitric oxide and endothelium

| Gene        | SNP · variant                 | Effect/risk allele | Functional consequence                                                                                                      | Cofactors                                                                            | Evidence (GWAS p, effect)                                                                                                                            | ClinVar · PharmGKB                                     |
|-------------|-------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>NOS3</b> | rs1799983 · Glu298Asp (G894T) | T (Asp298)         | Glu298Asp: Asp298 eNOS is more readily cleaved, giving less endothelial nitric oxide and less vasodilation during exercise. | BH4, FAD, FMN, heme, NADPH, Ca <sup>2+</sup> /calmodulin, L-arginine, O <sub>2</sub> | Cellular mechanism shown; MIXED clinical associations; PharmGKB 3. <b>Level: Moderate.</b> (Tesauro 2000)                                            | Benign · PharmGKB 3 (cardiovascular drugs)             |
| <b>NOS3</b> | rs2070744 · -786 T>C          | C (-786C)          | Promoter -786 T>C: the C allele lowers eNOS transcription.                                                                  | See NOS3 above                                                                       | Associated with coronary spasm; paradoxically more frequent in power athletes: MIXED evidence. <b>Level: Mixed.</b> (Gómez-Gallego 2009; Eynon 2012) | Benign; Protective · PharmGKB 3 (cardiovascular drugs) |
| <b>EDN1</b> | rs5370 · Lys198Asn            | T (Asn198)         | Lys198Asn in endothelin-1 (a vasoconstrictor): Asn198 is linked to higher endothelin and vascular reactivity.               | Endothelin-converting enzyme (Zn <sup>2+</sup> )                                     | MIXED studies; PharmGKB 3. <b>Level: Weak.</b>                                                                                                       | Benign · PharmGKB 3 (muraglitazar)                     |

### Capillaries

| Gene         | SNP · variant        | Effect/risk allele | Functional consequence                             | Cofactors                                | Evidence (GWAS p, effect)                                                        | ClinVar · PharmGKB                     |
|--------------|----------------------|--------------------|----------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------|
| <b>VEGFA</b> | rs2010963 · -634 G>C | C (-634C)          | VEGF-A -634 G>C promoter (muscle capillarisation). | Stimulated by HIF-1 $\alpha$ (see HIF1A) | Inconsistent results on VEGF levels and VO <sub>2</sub> max. <b>Level: Weak.</b> | Benign · PharmGKB 3 (cyclophosphamide) |

### Heart and renin–angiotensin system

| Gene         | SNP · variant         | Effect/risk allele | Functional consequence                                                                                                      | Cofactors                          | Evidence (GWAS p, effect)                                                                  | ClinVar · PharmGKB                                        |
|--------------|-----------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>ADRB1</b> | rs1801253 · Arg389Gly | G (Gly389)         | Arg389Gly in the cardiac $\beta$ 1 receptor: Arg389 gives stronger catecholamine responses (more cAMP).                     | Gs protein, ATP (adenylyl cyclase) | Beta-blocker response (PharmGKB 3); limited exercise data. <b>Level: Weak.</b>             | Benign; Association · PharmGKB 3 (muraglitazar)           |
| <b>AGTR1</b> | rs5186 · A1166C       | C (1166C)          | A1166C in the AT1 receptor 3' UTR: the C allele escapes miR-155, with more receptor and a stronger angiotensin II response. | —                                  | Vascular stiffness and cardiac adaptation: MIXED studies; PharmGKB 3. <b>Level: Mixed.</b> | Benign/likely benign · PharmGKB 3 (Ace Inhibitors, Plain) |

### Iron and oxygen transport

| Gene       | SNP · variant     | Effect/risk allele | Functional consequence                                                       | Cofactors | Evidence (GWAS p, effect)                                                                  | ClinVar · PharmGKB                                    |
|------------|-------------------|--------------------|------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>HFE</b> | rs1799945 · H63D  | G (63D)            | H63D in HFE: slightly higher iron absorption.                                | —         | Modest effect; relevant combined with C282Y. Ferritin target <100. <b>Level: Moderate.</b> | Conflicting classifications; Other · PharmGKB —       |
| <b>HFE</b> | rs1800562 · C282Y | A (282Y)           | C282Y in HFE: the main cause of hereditary haemochromatosis when homozygous. | —         | ClinVar pathogenic (low penetrance). <b>Level: Strong.</b>                                 | Pathogenic (low penetrance); Risk factor · PharmGKB — |

## 11. Glucose, fat and fuel

Muscle is the largest consumer of glucose after a meal; more muscle and more training improve insulin sensitivity. These genes affect insulin secretion, adiposity and fat mobilisation.

### Insulin and glucose

| Gene          | SNP · variant             | Effect/risk allele    | Functional consequence                                                                                            | Cofactors                                | Evidence (GWAS p, effect)                                                                                      | ClinVar · PharmGKB                                                                  |
|---------------|---------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>TCF7L2</b> | rs7903146 · intronic C>T  | T                     | Intronic in TCF7L2: the T allele lowers insulin secretion and beta-cell reserve.                                  | —                                        | Type 2 diabetes: OR 1.33 per allele, p $9 \times 10^{-471}$ . <b>Level: Strong.</b> (Grant 2006; Mahajan 2022) | Likely risk allele; Risk factor · PharmGKB 3 (sulfonylureas; sirolimus/ tacrolimus) |
| <b>MTNR1B</b> | rs10830963 · intronic C>G | G                     | Intronic in melatonin receptor MTNR1B: the G allele raises receptor expression in beta cells and fasting glucose. | Melatonin (ligand)                       | Fasting glucose and type 2 diabetes GWAS (p $< 10^{-50}$ ). <b>Level: Strong.</b> (GWAS Catalog)               | No record · PharmGKB —                                                              |
| <b>PPARG</b>  | rs1801282 · Pro12Ala      | G (Ala12, protective) | Pro12Ala in PPAR $\gamma$ (adipocytes): Ala12 improves insulin sensitivity.                                       | Ligands: fatty acids, thiazolidinediones | Type 2 diabetes: ~15–20% lower risk per Ala allele. <b>Level: Strong.</b> (Altshuler 2000)                     | Benign/likely benign · PharmGKB 3 (pioglitazone)                                    |

### Adiposity and appetite

| Gene        | SNP · variant               | Effect/risk allele | Functional consequence                                                                        | Cofactors                                           | Evidence (GWAS p, effect)                                                                                                                              | ClinVar · PharmGKB                             |
|-------------|-----------------------------|--------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>FTO</b>  | rs9939609 · intronic T>A    | A                  | Intronic in FTO (region regulating IRX3/IRX5 and appetite): the A allele increases adiposity. | Fe <sup>2+</sup> , 2-oxoglutarate (RNA demethylase) | Repeated GWAS (p $< 10^{-100}$ ); the effect is about 27% smaller in physically active people. <b>Level: Strong.</b> (Frayling 2007; Kilpeläinen 2011) | No record · PharmGKB 3 (interferon/ ribavirin) |
| <b>MC4R</b> | rs17782313 · intergenic T>C | C                  | Near MC4R (hypothalamic satiety): the C allele increases appetite and fat mass.               | —                                                   | Fat-mass GWAS: p $\sim 10^{-15}$ in large meta-analyses. <b>Level: Strong.</b> (Loos 2008)                                                             | No record · PharmGKB 3 (quetiapine)            |

### Lipolysis and thermogenesis

| Gene         | SNP · variant     | Effect/risk allele  | Functional consequence                                                    | Cofactors                    | Evidence (GWAS p, effect)                                               | ClinVar · PharmGKB                |
|--------------|-------------------|---------------------|---------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|-----------------------------------|
| <b>ADRB3</b> | rs4994 · Trp64Arg | G (Arg64)           | Trp64Arg in the $\beta_3$ receptor (adipose lipolysis and thermogenesis). | Gs protein, ATP              | Adiposity and insulin: MIXED results. <b>Level: Mixed.</b>              | Benign · PharmGKB 3 (olanzapine)  |
| <b>UCP2</b>  | rs660339 · A55V   | A (Val55, + strand) | A55V in uncoupling protein UCP2.                                          | Fatty acids; proton gradient | Energy efficiency and adiposity: inconsistent data. <b>Level: Weak.</b> | Benign/likely benign · PharmGKB — |

## 12. Brain, motivation and motor control

Early strength gains come mostly from the nervous system (recruitment and firing rate). The drive to exercise is also heritable: physical-activity GWAS find loci linked to dopamine and personality.

### Motor plasticity

| Gene        | SNP · variant     | Effect/risk allele | Functional consequence                                                                           | Cofactors                       | Evidence (GWAS p, effect)                                                            | ClinVar · PharmGKB                             |
|-------------|-------------------|--------------------|--------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------|
| <b>BDNF</b> | rs6265 · Val66Met | T (Met66)          | Val66Met in BDNF: Met66 reduces activity-dependent BDNF secretion (motor plasticity and memory). | Processing by furin and plasmin | Well-shown mechanism; MIXED effects on training. <b>Level: Moderate.</b> (Egan 2003) | Benign/likely benign · PharmGKB 3 (paroxetine) |

### Dopamine and reward

| Gene         | SNP · variant                 | Effect/risk allele | Functional consequence                                                                                             | Cofactors                                     | Evidence (GWAS p, effect)                                                                           | ClinVar · PharmGKB                   |
|--------------|-------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>COMT</b>  | rs4680 · Val158Met            | A (Met158)         | Val158Met in COMT: Met158 has ~3–4-fold lower activity (more prefrontal dopamine); Val/Val clears dopamine faster. | Mg <sup>2+</sup> , S-adenosylmethionine (SAM) | Functional and widely studied; MIXED effects on performance. <b>Level: Moderate.</b> (Lachman 1996) | Benign · PharmGKB 3 (antipsychotics) |
| <b>ANKK1</b> | rs1800497 · Taq1A (Glu713Lys) | A (A1)             | Taq1A (Glu713Lys) in ANKK1, linked to fewer striatal dopamine D2 receptors.                                        | —                                             | Reward and motivation: moderate evidence; PharmGKB 3. <b>Level: Moderate.</b> (Neville 2004)        | Benign · PharmGKB 3 (ethanol)        |

### Exercise behaviour

| Gene         | SNP · variant                           | Effect/risk allele       | Functional consequence                                        | Cofactors | Evidence (GWAS p, effect)                                                                                    | ClinVar · PharmGKB     |
|--------------|-----------------------------------------|--------------------------|---------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|------------------------|
| <b>CADM2</b> | rs62253088 · strenuous-sport GWAS locus | T (more strenuous sport) | CADM2 locus (synaptic adhesion; personality and risk-taking). | —         | Strenuous-sport GWAS: OR 1.05 per allele, p 1×10 <sup>-19</sup> . <b>Level: Moderate.</b> (Klimentidis 2018) | No record · PharmGKB — |



## 13. Hormones, mTOR and rapamycin

Free testosterone (not bound to SHBG) acts on the androgen receptor and raises protein synthesis and satellite cells. The receptor CAG repeat length (fewer repeats = more sensitive receptor) **cannot be read on a SNP chip**. Rapamycin inhibits mTORC1; its metabolism depends on CYP3A5/CYP3A4 and P-glycoprotein.

### Androgens and SHBG

| Gene   | SNP · variant                                      | Effect/risk allele                     | Functional consequence                                                                          | Cofactors                                    | Evidence (GWAS p, effect)                                                                                                                                                                                                           | ClinVar · PharmGKB                                                      |
|--------|----------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| AR     | CAG repeat (exon 1, X chromosome)                  | More repeats (less sensitive receptor) | The length of the glutamine repeat tunes androgen-receptor sensitivity to testosterone and DHT. | Testosterone/DHT; HSP90 chaperone            | Fewer repeats associated with more mass and strength in some studies but not others. <b>Level: Mixed.</b> (Guilherme 2021; Folland 2012). <b>Cannot be read on a SNP chip</b> (requires long-read sequencing or fragment analysis). | Expansion ≥38 repeats = spinal and bulbar muscular atrophy · PharmGKB — |
| SHBG   | rs12150660 · intronic (higher SHBG)                | T (higher SHBG)                        | Intronic in SHBG: the T allele raises SHBG and lowers the free testosterone fraction.           | Steroid binding; structural Ca <sup>2+</sup> | SHBG GWAS: +6.14 nmol/L per allele, p 1×10 <sup>-300</sup> . <b>Level: Strong.</b> (Harrison 2021; Ruth 2020)                                                                                                                       | No record · PharmGKB —                                                  |
| SHBG   | rs6258 · P156L                                     | T (156L, low SHBG)                     | P156L: the T allele reduces SHBG affinity and levels.                                           | See SHBG                                     | SHBG GWAS (the C allele is linked to higher SHBG). <b>Level: Strong.</b> (Ruth 2020)                                                                                                                                                | No record · PharmGKB —                                                  |
| SHBG   | rs727428 · 3' (T = more bioavailable testosterone) | T (more bioavailable testosterone)     | 3' of SHBG: the C allele raises SHBG; T gives more bioavailable testosterone.                   | See SHBG                                     | Bioavailable-testosterone GWAS: p 8×10 <sup>-309</sup> . <b>Level: Strong.</b> (Ruth 2020)                                                                                                                                          | No record · PharmGKB —                                                  |
| SHBG   | rs1799941 · -68 G>A (5' UTR)                       | A (higher SHBG)                        | -68 G>A in the SHBG 5' UTR: the A allele raises SHBG and total testosterone.                    | See SHBG                                     | SHBG and total testosterone GWAS. <b>Level: Strong.</b> (Ruth 2020)                                                                                                                                                                 | No record · PharmGKB —                                                  |
| FAM9B  | rs5934505 · X locus (low testosterone)             | T (low testosterone)                   | FAM9B locus on the X chromosome (men carry a single copy).                                      | —                                            | Low-testosterone GWAS: OR 1.57, p 2×10 <sup>-108</sup> ; T is the more frequent allele (~72%). <b>Level: Strong.</b> (GWAS Catalog)                                                                                                 | No record · PharmGKB —                                                  |
| SRD5A2 | rs523349 · V89L                                    | Leu89 (G on + strand)                  | V89L in type 2 5α-reductase: Leu89 reduces conversion of testosterone to DHT.                   | NADPH                                        | MIXED data; PharmGKB 3 (abiraterone). <b>Level: Mixed.</b>                                                                                                                                                                          | Benign · PharmGKB 3 (abiraterone)                                       |

### Vitamin D

| Gene | SNP · variant              | Effect/risk allele     | Functional consequence                                                                                   | Cofactors                                         | Evidence (GWAS p, effect)                                                                 | ClinVar · PharmGKB                                                                                   |
|------|----------------------------|------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| VDR  | rs2228570 · FokI (M1T)     | A (f allele, + strand) | FokI (M1T) in the vitamin D receptor: the f allele gives a protein 3 amino acids longer and less active. | 1.25(OH) <sub>2</sub> D, zinc (zinc fingers), RXR | Muscle strength in candidate-gene studies; MIXED. <b>Level: Mixed.</b> (Windelinckx 2007) | Uncertain significance; Conflicting classifications · PharmGKB 3 (peginterferon alpha-2b/ ribavirin) |
| GC   | rs2282679 · intronic (DBP) | G (lower 25-OH-D)      | Intronic in GC (vitamin D-binding protein).                                                              | —                                                 | 25-OH-D GWAS: p <10 <sup>-40</sup> . <b>Level: Strong.</b> (Wang 2010)                    | No record · PharmGKB —                                                                               |

### mTOR and longevity

| Gene  | SNP · variant                                   | Effect/risk allele        | Functional consequence                                                 | Cofactors                                               | Evidence (GWAS p, effect)                                                                                                     | ClinVar · PharmGKB  |
|-------|-------------------------------------------------|---------------------------|------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------|
| MTOR  | rs2295080 · promoter G>T                        | T/G (no consensus)        | MTOR promoter (central kinase of cell growth and protein synthesis).   | ATP, Mg <sup>2+</sup> ; amino acids (leucine), Rheb-GTP | GWAS: the T allele is linked to a lower insulin-sensitivity index (p 1×10 <sup>-8</sup> ). <b>Level: Weak.</b> (GWAS Catalog) | Benign · PharmGKB — |
| FOXO3 | rs2802292 · intronic T>G (G = longevity allele) | G (longevity, protective) | Intronic in FOXO3 (stress resistance, autophagy, muscle proteostasis). | —                                                       | Exceptional longevity replicated in several populations. <b>Level: Strong.</b> (Willcox 2008)                                 | Benign · PharmGKB — |

## Rapamycin pharmacogenomics

| Gene          | SNP · variant                   | Effect/risk allele    | Functional consequence                                                                                                     | Cofactors                   | Evidence (GWAS p, effect)                                                                                                                   | ClinVar · PharmGKB                                                                                         |
|---------------|---------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>CYP3A5</b> | rs776746 · *3 (6986A>G)         | C (*3, non-expresser) | CYP3A5*3: defective splicing; *1 (T allele) expresses the enzyme and clears its substrates (tacrolimus, sirolimus) faster. | Heme, NADPH, P450 reductase | CPIC guideline for tacrolimus (1A); sirolimus PharmGKB 3 (lower exposure in expressers). <b>Level: Strong.</b> (Birdwell 2015; te Loo 2023) | Benign; Association; Drug response; Risk factor · PharmGKB 1A tacrolimus (*3 haplotype; CPIC); 3 sirolimus |
| <b>CYP3A4</b> | rs35599367 · *22 (intron 6 C>T) | A (*22)               | CYP3A4*22: lower CYP3A4 expression.                                                                                        | Heme, NADPH                 | Higher exposure to substrates (tacrolimus, sirolimus); PharmGKB 3–4. <b>Level: Moderate.</b>                                                | No record · PharmGKB 3 (sirolimus, cyclosporine)                                                           |
| <b>ABCB1</b>  | rs1045642 · C3435T              | A (3435T)             | C3435T in ABCB1 (P-glycoprotein drug efflux pump).                                                                         | ATP, Mg <sup>2+</sup>       | Many drugs, MIXED results; sirolimus PharmGKB 3. <b>Level: Mixed.</b>                                                                       | Benign · PharmGKB 3 (sirolimus)                                                                            |

## 14. Summary table: categories, genes, cofactors and targets

Supplementation or lifestyle targets with trial or meta-analysis evidence, linked to the biology of each category. They are not individual recommendations: they should be discussed with a treating physician.

| Category                          | Genes                                                                | Cofactors                                                                                                                     | Supplement / lifestyle targets                                                                                                                                 |
|-----------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Fibre, power                   | ACTN3, AMPD1, ACE, AGT, HIF1A, PPARA, IL6                            | Zn <sup>2+</sup> (AMPD1, ACE); Fe <sup>2+</sup> , 2-oxoglutarate and vitamin C (HIF hydroxylases)                             | Creatine (phosphagen system; Kreider 2017); dietary nitrate for power (Coggan 2021); caffeine (Guest 2021); power training                                     |
| 2. Myofibrillar hypertrophy       | IGF1, IGF1R, TRHR, MSTN, FST, MTOR                                   | Amino acids (leucine), adequate energy, zinc; ATP/Mg <sup>2+</sup> (kinases)                                                  | Adequate protein over 3–4 meals (Morton 2018); creatine (Desai 2024); omega-3 (Smith 2011); avoid a large energy deficit (Murphy 2022)                         |
| 3. Sarcoplasmic, energy           | PPARGC1A, TFAM, SLC16A1, CKM                                         | Mg <sup>2+</sup> -ATP, creatine/phosphocreatine; iron and copper (respiratory chain)                                          | Creatine; magnesium at nutritional dose; iron only if deficient                                                                                                |
| 4. Strength, trainability         | CNTF, IL15RA, PPARG, ACSL1, loci GWAS                                | ATP, coenzyme A, Mg <sup>2+</sup> (ACSL1)                                                                                     | Progressive training is the main modulator (Schoenfeld 2017); vitamin D if low (Beaudart 2014)                                                                 |
| 5. Damage, recovery               | MYLK, LILRB5, TNF, CCL2, SOD2, SLC01B1                               | Ca <sup>2+</sup> /calmodulin (MYLK); Mn <sup>2+</sup> (SOD2)                                                                  | Omega-3; sleep; avoid antioxidant megadoses around training (Ristow 2009; Paulsen 2014a; Paulsen 2014b)                                                        |
| 6. Tendons                        | COL5A1, COL1A1, COL3A1, GDF5, MMP3, MMP7                             | Vitamin C, Fe <sup>2+</sup> , 2-oxoglutarate (hydroxylases); copper (lysyl oxidase); Zn <sup>2+</sup> /Ca <sup>2+</sup> (MMP) | Gelatin or collagen with vitamin C 30–60 min before loading (Shaw 2017); slow load progression; avoid fluoroquinolones when an alternative exists (Alves 2019) |
| 7. Cardiovascular, O <sub>2</sub> | NOS3, EDN1, VEGFA, ADRB1, AGTR1, HFE                                 | BH4, FAD/FMN, heme, NADPH, Ca <sup>2+</sup> /calmodulin, L-arginine (eNOS); iron (haemoglobin)                                | Dietary nitrate (leafy greens, beetroot); aerobic training; iron only if low                                                                                   |
| 8. Glucose, fat                   | TCF7L2, MTNR1B, PPARG, FTO, MC4R, ADRB3, UCP2                        | Melatonin (MTNR1B ligand); Fe <sup>2+</sup> /2-oxoglutarate (FTO)                                                             | More fibre; strength and aerobic training (activity attenuates FTO; Kilpeläinen 2011)                                                                          |
| 9. Brain, motivation              | BDNF, COMT, ANKK1, CADM2                                             | Mg <sup>2+</sup> , S-adenosylmethionine (COMT)                                                                                | Regular exercise (raises BDNF); sleep; moderate caffeine (Guest 2021)                                                                                          |
| 10. Hormones, mTOR, PGx           | AR, SHBG, FAM9B, SRD5A2, VDR, GC, MTOR, FOXO3, CYP3A5, CYP3A4, ABCB1 | NADPH (SRD5A2, CYP); 1.25(OH) <sub>2</sub> D and zinc (VDR); heme (CYP)                                                       | Vitamin D if low; zinc at nutritional dose; drugs: informational only, with the prescribing physician                                                          |

## 15. SNP lookup table (GRCh38)

Positions on the GRCh38 assembly (1-based). Ref/Alt = reference and alternative alleles on the + strand. To look up in a VCF, use the rsID first and, if it is not annotated, the position.

| rsID       | Gene     | Chr | GRCh38 position | Ref/Alt | Published effect allele                   | Cat. |
|------------|----------|-----|-----------------|---------|-------------------------------------------|------|
| rs1815739  | ACTN3    | 11  | 66 560 624      | C/T     | T (X, stop codon)                         | 1    |
| rs17602729 | AMPD1    | 1   | 114 693 436     | G/A     | A (34T, stop codon)                       | 1    |
| rs558129   | GALNTL6  | 4   | 171 829 960     | G/A     | T (gene strand)                           | 1    |
| rs4343     | ACE      | 17  | 63 488 670      | G/A     | G (tags the D allele)                     | 1    |
| rs699      | AGT      | 1   | 230 710 048     | A/G     | G (235T)                                  | 1    |
| rs11549465 | HIF1A    | 14  | 61 740 839      | C/T     | T (Ser582, advantage)                     | 1    |
| rs4253778  | PPARA    | 22  | 46 234 737      | G/C     | C (power)                                 | 1    |
| rs1800795  | IL6      | 7   | 22 727 026      | C/G     | G (-174G)                                 | 1    |
| rs35767    | IGF1     | 12  | 102 481 791     | A/G     | T in the gene (A on + strand)             | 2    |
| rs2229765  | IGF1R    | 15  | 98 934 996      | G/A     | A                                         | 2    |
| rs7832552  | TRHR     | 8   | 109 103 447     | C/T     | T (more lean mass)                        | 2    |
| rs1805086  | MSTN     | 2   | 190 060 351     | T/C     | C (153R)                                  | 2    |
| rs3797297  | FST      | 5   | 53 481 826      | G/T     | T                                         | 2    |
| rs8192678  | PPARGC1A | 4   | 23 814 039      | C/T     | T (Ser482)                                | 3    |
| rs1937     | TFAM     | 10  | 58 385 582      | G/C     | C (Thr12)                                 | 3    |
| rs1049434  | SLC16A1  | 1   | 112 913 924     | A/T     | T (gene strand, 1470T)                    | 3    |
| rs8111989  | CKM      | 19  | 45 305 950      | T/C     | A/G (NcoI; no consensus risk allele)      | 3    |
| rs1800169  | CNTF     | 11  | 58 624 028      | G/A     | A (null allele)                           | 4    |
| rs2296135  | IL15RA   | 10  | 5 952 731       | A/C     | A (more hypertrophy)                      | 4    |
| rs2267668  | PPARD    | 6   | 35 410 145      | G/A     | G (lower response)                        | 4    |
| rs6552828  | ACSL1    | 4   | 184 804 262     | A/G     | A (+ strand)                              | 4    |
| rs958685   | TGFA     | 2   | 70 476 715      | C/A     | A (more strength)                         | 4    |
| rs72979233 | POLD3    | 11  | 74 644 478      | A/G     | A (more strength)                         | 4    |
| rs2288278  | HOXB3    | 17  | 48 583 930      | G/A     | A (more strength)                         | 4    |
| rs4926611  | GLIS1    | 1   | 53 646 060      | T/C     | C (more strength)                         | 4    |
| rs6687430  | PEX14    | 1   | 10 573 188      | G/A     | G (more strength)                         | 4    |
| rs80103986 | KANSL1   | 17  | 46 147 574      | G/A     | A (more strength)                         | 4    |
| rs10186876 | LRPPRC   | 2   | 44 014 016      | A/G     | A (more strength)                         | 4    |
| rs28497577 | MYLK     | 3   | 123 793 780     | G/T     | A in the gene (37885A)                    | 5    |
| rs2700352  | MYLK     | 3   | 123 831 616     | G/A     | T in the gene (49T)                       | 5    |
| rs12975366 | LILRB5   | 19  | 54 255 498      | T/C     | T (Asp247)                                | 5    |
| rs1800629  | TNF      | 6   | 31 575 254      | G/A     | A (-308A)                                 | 5    |
| rs1024611  | CCL2     | 17  | 34 252 769      | A/G     | G (-2518G)                                | 5    |
| rs4880     | SOD2     | 6   | 159 692 840     | A/G     | G (Ala16)                                 | 5    |
| rs4149056  | SLCO1B1  | 12  | 21 178 615      | T/C     | C (Ala174)                                | 5    |
| rs12722    | COL5A1   | 9   | 134 842 570     | C/T     | T                                         | 6    |
| rs1800012  | COL1A1   | 17  | 50 200 388      | C/A     | T in the gene ("s" allele, A on + strand) | 6    |
| rs1800255  | COL3A1   | 2   | 188 999 354     | G/A     | A (698T)                                  | 6    |
| rs143383   | GDF5     | 20  | 35 438 203      | G/A     | T in the gene (A on + strand)             | 6    |
| rs679620   | MMP3     | 11  | 102 842 889     | T/C     | E45K (no consensus)                       | 6    |
| rs1249269  | MPP7     | 10  | 28 331 214      | T/C     | T                                         | 6    |
| rs1799983  | NOS3     | 7   | 150 999 023     | T/G     | T (Asp298)                                | 7    |
| rs2070744  | NOS3     | 7   | 150 992 991     | C/T     | C (-786C)                                 | 7    |
| rs5370     | EDN1     | 6   | 12 296 022      | G/T     | T (Asn198)                                | 7    |
| rs2010963  | VEGFA    | 6   | 43 770 613      | C/G     | C (-634C)                                 | 7    |

|            |        |    |             |     |                                    |    |
|------------|--------|----|-------------|-----|------------------------------------|----|
| rs1801253  | ADRB1  | 10 | 114 045 297 | G/C | G (Gly389)                         | 7  |
| rs5186     | AGTR1  | 3  | 148 742 201 | A/C | C (1166C)                          | 7  |
| rs1799945  | HFE    | 6  | 26 090 951  | C/G | G (63D)                            | 7  |
| rs1800562  | HFE    | 6  | 26 092 913  | G/A | A (282Y)                           | 7  |
| rs7903146  | TCF7L2 | 10 | 112 998 590 | C/T | T                                  | 8  |
| rs10830963 | MTNR1B | 11 | 92 975 544  | C/G | G                                  | 8  |
| rs1801282  | PPARG  | 3  | 12 351 626  | C/G | G (Ala12, protective)              | 8  |
| rs9939609  | FTO    | 16 | 53 786 615  | T/A | A                                  | 8  |
| rs17782313 | MC4R   | 18 | 60 183 864  | T/C | C                                  | 8  |
| rs4994     | ADRB3  | 8  | 37 966 280  | A/G | G (Arg64)                          | 8  |
| rs660339   | UCP2   | 11 | 73 978 059  | G/A | A (Val55, + strand)                | 8  |
| rs6265     | BDNF   | 11 | 27 658 369  | C/T | T (Met66)                          | 9  |
| rs4680     | COMT   | 22 | 19 963 748  | G/A | A (Met158)                         | 9  |
| rs1800497  | ANKK1  | 11 | 113 400 106 | G/A | A (A1)                             | 9  |
| rs62253088 | CADM2  | 3  | 85 351 651  | T/C | T (more strenuous sport)           | 9  |
| rs12150660 | SHBG   | 17 | 7 618 597   | G/T | T (higher SHBG)                    | 10 |
| rs6258     | SHBG   | 17 | 7 631 360   | C/T | T (156L, low SHBG)                 | 10 |
| rs727428   | SHBG   | 17 | 7 634 474   | T/C | T (more bioavailable testosterone) | 10 |
| rs1799941  | SHBG   | 17 | 7 630 105   | G/A | A (higher SHBG)                    | 10 |
| rs5934505  | FAM9B  | X  | 8 945 785   | T/C | T (low testosterone)               | 10 |
| rs523349   | SRD5A2 | 2  | 31 580 636  | G/C | Leu89 (G on + strand)              | 10 |
| rs2228570  | VDR    | 12 | 47 879 112  | A/G | A (f allele, + strand)             | 10 |
| rs2282679  | GC     | 4  | 71 742 666  | T/G | G (lower 25-OH-D)                  | 10 |
| rs2295080  | MTOR   | 1  | 11 262 571  | G/T | T/G (no consensus)                 | 10 |
| rs2802292  | FOXO3  | 6  | 108 587 315 | G/T | G (longevity, protective)          | 10 |
| rs776746   | CYP3A5 | 7  | 99 672 916  | T/C | C (*3, non-expresser)              | 10 |
| rs35599367 | CYP3A4 | 7  | 99 768 693  | G/A | A (*22)                            | 10 |
| rs1045642  | ABCB1  | 7  | 87 509 329  | A/G | A (3435T)                          | 10 |

AR (CAG repeat, approx. chrX:67,545,317, exon 1) and ACE I/D (intron-16 Alu insertion) are not SNPs: they need dedicated methods.

## 16. Controversial evidence and limits

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- **ACTN3 R577X**: the most replicated performance gene in elite athletes (Yang 2003; Ma 2013), but it explains little in non-athletes; XX does not prevent strength gains with training (Clarkson 2005).
- **ACE I/D**: 25 years of studies with contradictory results (Puthucherry 2011). D is associated with higher enzyme activity, but the link to strength and power is inconsistent.
- **Candidate genes in general** (PPARA, HIF1A, IL6, AGT, CNTF, IL15RA, MYLK, COL5A1, COL1A1): small samples and false-positive risk; few survive in GWAS (Semenova 2023; Bray 2009).
- **Frequency/effect data**: for many of these variants, frequency/effect data are weak or absent in the reference population outside the original samples.
- **Not read by chips**: AR CAG repeat, ACE insertion/deletion, rare large-effect variants (for example, myostatin loss of function), structural variants and heteroplasmic mitochondrial DNA variants.
- **Imputation**: imputed genotypes should have quality ( $R^2/DR2$ )  $\geq 0.8$  to be used; below that they should not guide decisions.

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